HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Engineered a model-agnostic data extraction layer using **Python** and **Revit API**, ensuring full element-level traceability for downstream processing.
- Developed an **AI compliance engine** in **Python** that identifies duplicated, overlapping, and suppressed model elements, reducing data quality issues by approximately **20%** before measurement.
- Built an automated quantity take-off module with **AI classification** for fittings and accessories, generating Standard-Method-of-Measurement-aligned Bills of Quantities.
- Designed and implemented a **BIM-AI decision dashboard** integrating quantities, cost, and embodied carbon data, featuring revision differencing for scenario comparison.
- Deployed developed modules on live construction projects, benchmarking performance against manual processes and documenting findings for project evaluation gates.
- Trained company staff on new BIM and AI tools, producing comprehensive user guides and operational documentation to facilitate platform adoption.
- Containerized AI and data processing modules using **Docker** and deployed them to an **AWS** environment, configuring basic resource limits for predictable performance.

SKILLS

BIM & Digital Construction – Revit API, IFC, open BIM, Dynamo, quantity take-off, 5D BIM, BIM design, offsite prefabrication

AI & Machine Learning – AI compliance checking, AI classification, AI cost management, retrieval-augmented rate suggestion, large-language-model development

Programming & Scripting – Python, C# (desirable), shell scripting

Data & Analytics – data-quality checking, cost management, embodied carbon, data extraction, data processing

Cloud & DevOps – AWS, Docker

Project Management & Collaboration – Agile methodologies, Scrum, change-control management, user guides, documentation, stakeholder communication

PROJECTS

AI-Driven BIM Model Compliance Checker

- Developed a **Python**-based engine to parse **Revit** and **IFC** files, identifying and flagging non-compliant model elements based on defined quality rules.
- Implemented logic to detect duplicated, overlapping, and suppressed elements, improving model data integrity by an estimated **15%** in pilot projects.
- Integrated a user-friendly interface for engineers to review flagged issues, enabling an engineer-approve-in-the-loop workflow for automated detailing.

Automated Quantity Take-Off and Costing Integration

- Engineered a system to automatically extract quantities from BIM models, classifying fittings and accessories using **Python** and AI techniques.
- Integrated the take-off data with a simulated costing system, enabling AI-driven rate suggestions and the assembly of preliminary cost plans.
- Designed the data flow for model-linked cost control, supporting cash flow and earned value tracking for project management.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)